AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

# T103/TM103
# Computer Organization and Architecture

## Final Examination

## Spring Semester 2024/2025

## Date: 12 /June/2025

| Number of Exam Pages: 4 (including this cover sheet) | Time Allowed: 2 Hours |
|---|---|

**Instructions:**

- Total Marks: **100**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Programmable Calculators** are **not** allowed.



CamScanner

## Part I: Choose the most correct answer [20 marks]

This part consists of **10 questions** carrying a **weight of two marks each**.

**Q-1:** Answer all the following questions by choosing the most correct statement.

1. The CPU component responsible for executing the instruction and placing the results in registers is:
   a) D-Flip Flop.
   b) Register.
   c) Control Unit.
   d) Arithmetic Logic Unit.
2. If we have an 8M X 64-bit memory module (Byte addressable), then the needed number of address lines is:
   a) 28.
   b) 23
   c) 26
   d) 16.
3. Which of the following is not true about Static RAM:
   a) Needs continuous electricity.
   b) Used as Cache memory.
   c) Very cheap compared to DRAM
   d) Both a and c.
4. The register that holds the address of the next instruction to be executed is:
   a) Status.
   b) Instruction.
   c) Program Counter.
   d) Accumulator.
5. An indirect addressing mode means that:
   a) The address of the operand is in the address stated in the instruction.
   b) The address of the operand is explicitly stated in the instruction.
   c) The address of the operand is to be found in the ALU.
   d) None of the choices.
6. A counter that can have 32 different states, needs:
   a) 64 Flip-Flops
   b) 8 Flip-Flops.
   c) 4 Flip-Flops.
   d) None of the choices.
7. If a function F is equal to 1 for the expression (x+y)', then:
   a) Both inputs are 1.
   b) Only one of the inputs is 0.
   c) Only one of the inputs is 1.
   d) Both inputs are 0.
8. ..... is an intelligent serial interface that supports "plug and play":
   a) Universal Serial Bus.
   b) Parallel Port.
   c) Serial port.
   d) None of the choices.
9. If a carry-out of the leftmost bit occurs in an addition of two signed numbers using 1's complement, then:
   a) It will be discarded.
   b) This means that we surely have an erroneous result.
   c) It will be added to the sum.
   d) None of the choices.
10. Which of the following is a characteristic of EPROM?
    a) It can be programmed once.
    b) It is volatile.
    c) Erasure is performed using a special tool.
    d) Both b and c.



CamScanner

## Part II: Short essay questions [30 marks]

This part consists of **five questions** carrying a **WEIGHT OF 6 marks each**.
You should answer **all** questions.

**Q-2:** What do the following acronyms stand for: EEPROM, CPU, MAR?

**Q-3:** Write the RTN of **Add y** and **Load y**.

**Q-4:** Given the K-Map diagram below, answer the following questions:

a) Find the Sum of Product form of the Function F.
b) Deduce the simplified form of the function F.

| | b'c' | b'c | bc | bc' |
|---|---|---|---|---|
| a' | 1 | 0 | 0 | 1 |
| a | 0 | 0 | 1 | 1 |

**Q-5:** Using MARIE ISA, find the equivalent assembly codes of the following binary instructions.

a) **1000** 0100 **0000** 0000.
b) **0101** 0000 **0000** 0000.

**Q-6:** Draw the characteristic table of the **S-R Flip-Flop** and its Logical diagram using NOR gates.

---

## Part III: Numerical problems [50 marks]

This part consists of **four questions** carrying a weight of **50 marks**.
You should **answer all questions**.

**Q-7:** Suppose that you have the following pseudocode excerpt: **[10 marks]**

```
Input a number
Assign this number to x
Input a number
Assign this number to y
Jump to the output line
Let AC= x
Display AC value
Stop
```

a) Write a MARIE assembly code to perform the above pseudocode excerpt, including the needed labels and directives.
b) What will be the output if the entered numbers are 6 and 9 respectively?

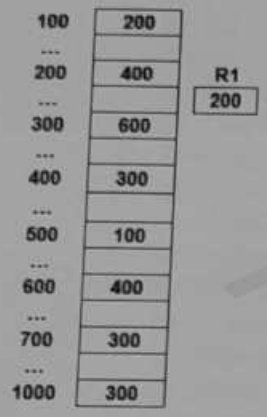


CamScanner

**Q-8:** Suppose we have the following code: **[12 marks]**

**Add 400**
**Load 200**

Knowing that **AC=600** before executing the two instruction, and the memory and register R1 containing the values below, determine the actual value that will be found in the accumulator **AC** after executing each instruction in each mode, and fill in the table below: Note that for each addressing mode the value of AC is initially 600




| Mode | Value in AC after Add 400 | Value in AC after Load 200 |
|---|---|---|
| Immediate | | |
| Direct | | |
| Indirect | | |
| Indexed | | |

**Q-9:** Assume that we have the following Direct Mapped Cache representation: **[14 marks]**

Direct Mapped

| 7 | 11 | 6 |
|---|---|---|

Answer the following questions based on the above data:

**a-** What is the size of the main memory?
**b-** What is the number of blocks in the cache memory?
**c-** How many blocks are there in the main memory?

Draw the format of the memory address as seen by the cache, showing the fields as well as their sizes, for the following scheme:

**d- 16-way** set associative cache mapping scheme

**Q-10:** Answer the following questions: **[14 marks]**

a) Find the values of (-38) and (-98) in 8-bits binary representation using **C1** representation.
b) Find the sum of (-38) and (-98) in 8-bits binary representation using **C1** representation.
c) What can you say about the result? Discuss.

***End of Questions***



CamScanner